

Heisenberg picture

For a time-independent Hamiltonian,

$$U = \exp\left[-\frac{i}{\hbar}(t - t_0)H\right].$$

States:

$$|\psi\rangle_H = U|\psi(t)\rangle_S.$$

But since

$$[H, H] = 0,$$

Then

$$\exp\left[-\frac{i}{\hbar}(t - t_0)H\right] \exp\left[\frac{i}{\hbar}(t - t_0)H\right] = \mathbb{I}.$$

Using

$$e^A e^B = e^{A+B} \iff [A, B] = 0$$

(the Campbell–Baker–Hausdorff formula), one obtains

$$|\psi\rangle_H = |\psi_0\rangle, \quad \text{NO TIME DEPENDENCY.}$$

Operator:

$$A_H = UAU^\dagger.$$

Thus

$A_H(t)$ picks up a time dependency.

Differential eq.?

$$i\hbar \partial_t A_H(t) = i\hbar \left[(\partial_t U)AU^\dagger + UA(\partial_t U^\dagger) \right].$$

but

$$\partial_t U = -\frac{i}{\hbar}HU, \quad \partial_t U^\dagger = \left(-\frac{i}{\hbar}HU\right)^\dagger = \frac{i}{\hbar}U^\dagger H \quad (H^\dagger = H),$$

then

$$i\hbar \partial_t A_H(t) = -HA_H(t) + A_H(t)H = [A_H(t), H].$$

d) Interaction picture

Let us assume that H can be decomposed into

$$H = H_0 + V,$$

where H_0 is the Hamiltonian part and V is the interaction part. Then we choose

$$U = \exp\left[-\frac{i}{\hbar}(t - t_0)H_0\right].$$

States:

$$|\psi\rangle_I = U|\psi(t)\rangle_S = \exp\left[-\frac{i}{\hbar}(t - t_0)H_0\right] |\psi(t)\rangle_S.$$

Interaction picture *S picture*

$$|\psi\rangle_I \implies \text{TIME DEPENDENCY.}$$

Operator:

$$A_I = U(t)AU^\dagger(t),$$

$$A_I(t) \longleftarrow \text{TIME DEPENDENCY.}$$

We can show that (check) $A_I(t)$ is solution of

$$i\hbar \partial_t A_I(t) = [A_I(t), H_0].$$

The time evolution of A_I is the one of a free object.

$\implies$ Quantum Oscillator (Phys-...)

Quantum oscillator

the Classical 1D quantum oscillator:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2, \quad [x, p] = i\hbar.$$

Introducing dimensionless (dimensionless) variables

Introduce

$$X = \sqrt{\frac{m\omega}{\hbar}} x, \quad P = \frac{1}{\sqrt{m\hbar\omega}} p.$$

Then

$$H = \frac{1}{2}\hbar\omega(X^2 + P^2), \quad [X, P] = i.$$

Ladder operators

We introduce the ladder operators

$$a = \frac{1}{\sqrt{2}}(X + iP), \quad a^\dagger = \frac{1}{\sqrt{2}}(X - iP),$$

with

$$[a, a^\dagger] = 1.$$

Then

$$H = \hbar\omega \left(\frac{1}{2} + a^\dagger a \right).$$

We introduce the Hermitian operator

$$N = a^\dagger a.$$

It satisfies

$$[N, a] = -a, \quad [N, a^\dagger] = a^\dagger, \quad [N, H] = 0.$$

There is a common set of eigenstates $\{|n\rangle\}$:

$$N|n\rangle = n|n\rangle, \quad H|n\rangle = E_n|n\rangle.$$

It is equivalent to use $\{E_n\}$ to describe the states.

Action of the operator a on a $|n\rangle$ state

$$[N, a]|n\rangle = N(a|n\rangle) - a(N|n\rangle) = N(a|n\rangle) - n(a|n\rangle) = -a|n\rangle.$$

Therefore

$$N(a|n\rangle) = (n-1)(a|n\rangle).$$

Thus $a|n\rangle$ is an eigenstate of N with eigenvalue $(n-1)$:

$$a|n\rangle = \alpha|n-1\rangle.$$

The normalization factor is determined by

$$|\alpha|^2 = \langle n-1|\alpha^*\alpha|n-1\rangle = \langle n|a^\dagger a|n\rangle = \langle n|N|n\rangle = n.$$

For $n \geq 0$ real, choose

$$\alpha = \sqrt{n}.$$

Hence

$$a|n\rangle = \sqrt{n}|n-1\rangle.$$

The vacuum state satisfies

$$a|0\rangle = 0.$$

The annihilation operator removes one unit of N , i.e. one quantum of energy $\hbar\omega$.

Creation operator

For the same reason,

$$a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle.$$

This is the creation operator. Starting from the vacuum,

$$\frac{1}{\sqrt{n!}}(a^\dagger)^n|0\rangle = |n\rangle.$$

Chapter 2: Classical Field Theory

Continuous systems — classical fields

A field theory is a theory with a continuum of variables instead of a set of discrete variables.

From a discrete to a continuous formulation

Problem set up:

Consider an elastic string: an infinite chain of point masses attached with identical massless springs. The set

$$\{\varphi_i, i \in \{1, N\}\}$$

is the set of discrete variables representing a possible state of the system.

We will assume short-range interaction. The Hamiltonian is

$$H = \sum_i \left[\frac{1}{2} m \dot{\varphi}_i^2 + \frac{1}{2} k (\varphi_{i+1} - \varphi_i)^2 \right].$$

It is useful to factor out a :

$$H = a \sum_i \left[\frac{1}{2} \frac{m}{a} \dot{\varphi}_i^2 + \frac{1}{2} k a \left(\frac{\varphi_{i+1} - \varphi_i}{a} \right)^2 \right].$$

Equation of motion: Euler–Lagrange equation.

The corresponding Lagrangian is

$$L = a \sum_i \left[\frac{1}{2} \frac{m}{a} \dot{\varphi}_i^2 - \frac{1}{2} k a \left(\frac{\varphi_{i+1} - \varphi_i}{a} \right)^2 \right].$$

The Euler–Lagrange equation is

$$\frac{\partial L}{\partial \varphi_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\varphi}_i} = 0.$$

Show that

$$Q = -\frac{m}{a} \ddot{\varphi}_i - k a \frac{2\varphi_i - \varphi_{i+1} - \varphi_{i-1}}{a^2}.$$

One obtains

$$\frac{m}{a} \ddot{\varphi}_i - k a \frac{2\varphi_i - \varphi_{i+1} - \varphi_{i-1}}{a^2} = 0.$$

Going to the continuum

This amounts to taking $a \rightarrow 0$ in a specific way:

$$\frac{m}{a} \longrightarrow \mu, \quad k a \longrightarrow \chi, \quad i a \longrightarrow x.$$

Here μ is the mass density of the string, χ is Young’s modulus of the string, and x is the position of the string.

Displacement field: mapping of a space-time point is a physical quantity.

The displacement field is the map

$$\varphi_i(t) \longrightarrow \varphi(x, t).$$

The discrete set of variables becomes a continuous set of variables; a point in space-time is a physical quantity.

The difference becomes

$$\varphi_{i+1}(t) - \varphi_i(t) \longrightarrow \varphi(x + a, t) - \varphi(x, t) = \varphi(x, t) + \frac{\partial \varphi}{\partial x} a + O(a^2).$$

In particular,

$$\varphi(x + a, t) - \varphi(x, t) \simeq \frac{\partial \varphi}{\partial x} a.$$

Also,

$$a \sum_i \longrightarrow \int_0^L dx.$$

Thus, the Lagrangian is

$$L = \int_0^L dx \left[\frac{1}{2} \mu (\partial_t \varphi)^2 - \frac{1}{2} \chi (\partial_x \varphi)^2 \right].$$

The Lagrangian density is

$$\mathcal{L} = \frac{1}{2} \mu (\partial_t \varphi)^2 - \frac{1}{2} \chi (\partial_x \varphi)^2.$$

We have also the action of this system:

$$S = \int_{t_1}^{t_2} dt \int_0^L dx \mathcal{L}.$$

Equivalently,

$$S = \int_{t_1}^{t_2} dt \int_0^L dx \mathcal{L}(\partial_t \varphi, \partial_x \varphi).$$

By taking the continuum limit of the discrete equation of motion, one gets

$$-\mu \left(\frac{\partial \varphi}{\partial t} \right)^2 + \chi \left(\frac{\partial \varphi}{\partial x} \right)^2 = 0.$$

Comments

1. In general,

$$\mathcal{L} = \mathcal{L}(\partial_t \varphi, \partial_x \varphi; t).$$

2. A discrete system with short-range interaction becomes, in the continuum limit, a Lagrangian density depending on first-order derivatives of the field with respect to time and space.
3. The corresponding action is local because S is an integral over time and space of a Lagrangian density which depends on the values of the field and its derivatives at one point. Moreover, the Lagrangian density is a polynomial.

Remarks

- The Lagrangian formalism will be mainly used because it treats time and space on the same footing (relativity).
- The Hamiltonian makes time explicit (space is disentangled).
- The field $\varphi(x, t)$ has continuous variables; it is a dynamical variable (continuous).